

Flight dynamics and Aerodynamic modelling of Cessna 550

Muhammed Billal Noor

Task 1 Aerodynamic model:

For the Cessna 550 the aerodynamic model was created using the provided aerodynamic coefficients. These were made from the lift, pitching moment, drag, rolling moment, yawing and side force and then implemented as angle of attack functions and control surface deflections.

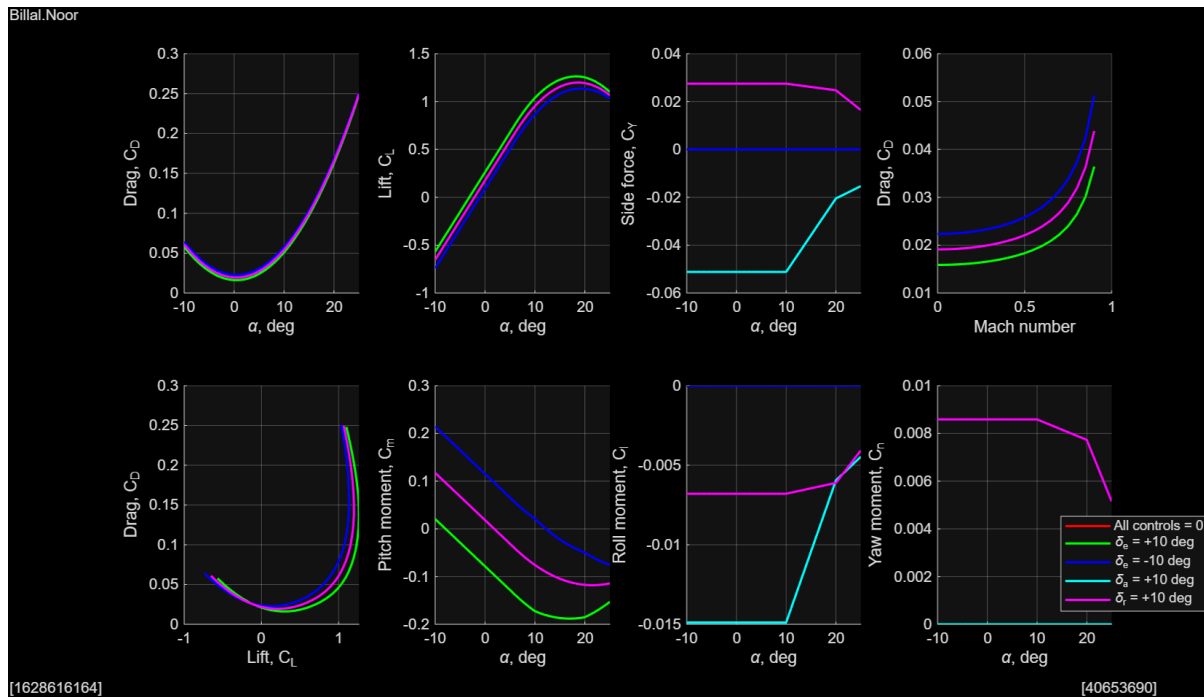


Figure 1: Aerodynamic coefficients as angle of attack functions for control surface deflections.

The aerodynamic coefficients show the influence of how control surfaces can affect aircraft behaviour. The elevator deflection affected the lift and pitching moment since the positive deflection increases lift and causes a pitching moment with the aircraft's nose down.

The aileron deflection affected the rolling moment and the rudder deflection affected the yawing moment and side force. When the angles of attack increase the behaviour becomes non linear since the effectiveness of control is reduced. This makes it consistent with realistic aerodynamic characteristics.

The aircraft model was implemented in MATLAB using the scripts. The aerodynamic model was defined in `c550aero.m` and the aircraft parameters were initialised in `c550init.m`.

Task 2 Trimming in SSLF:

The cessna 550 aircraft was trimmed at a altitude of 4000m with varying velocities.

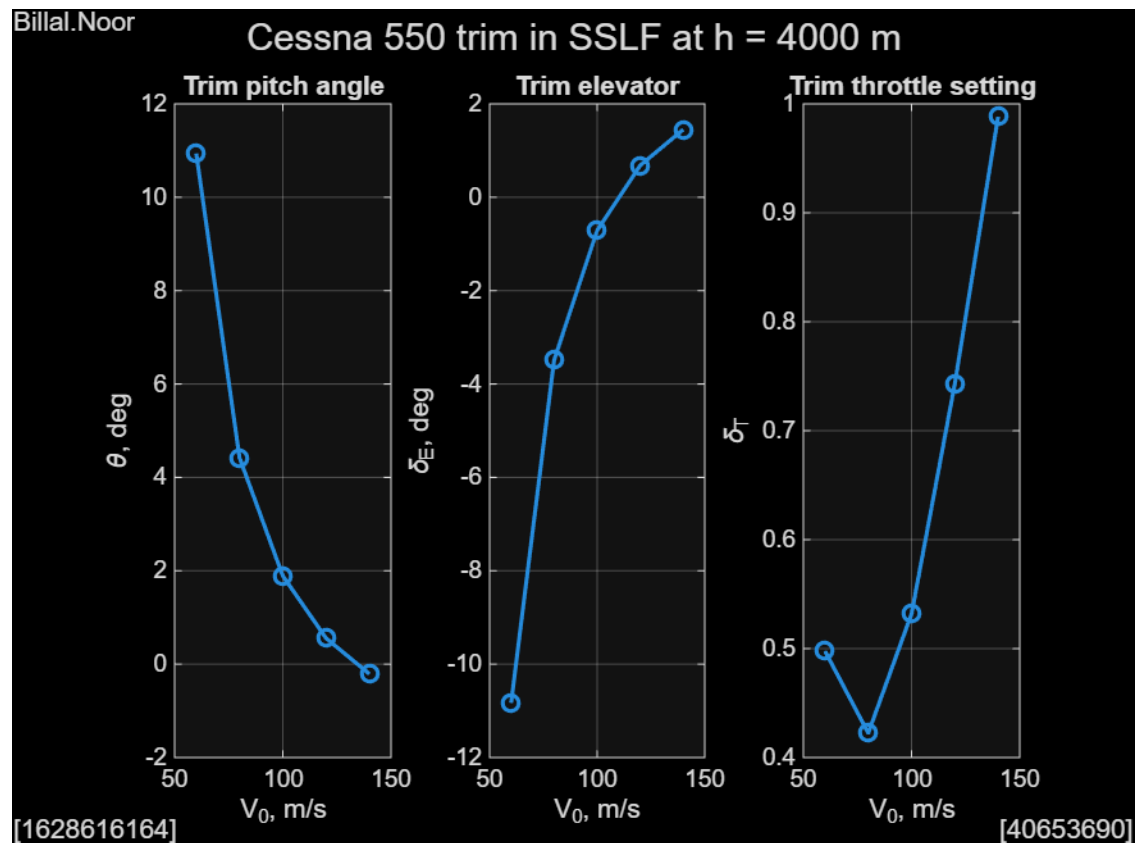


Figure 2: Cessna 550 trim plots.

V_0 (m/s)	ExitFlag	θ (deg)	δ_e (deg)	Throttle	Cost
60	1	10.938	-10.830	0.4977	5.76e-13
80	1	4.419	-3.480	0.4225	7.73e-13
100	1	1.890	-0.711	0.5319	3.17e-12
120	1	0.555	0.682	0.7431	5.45e-12
140	1	-0.210	1.450	0.9883	5.49e-12
160	0	—	—	—	7.13e-02
180	0	—	—	—	3.79

As the velocity increased the pitch angle decreases which indicated that lower angles of attack were needed. the elevator deflection angle moves from negative to positive

values and the throttle increases due to drag.

The trim failure at higher speeds (160 and 180) show that there could have been convergence issues or there were limitations in control as thrust was already near maximum output in 140 (0.98).

Aircraft trim occurs when all the forces and moments are balanced

$$\dot{x} = 0$$

this keeps the flight conditions steady with a constant velocity.

Trimming was completed using Model_Trim_SSLF.m which optimizes the elevator deflection, throttle and pitch angle the residual forces and moments are kept to a minimum until equilibrium is reached.

The algorithm in the Model_Trim_SSLF.m script optimises the control variables (elevator deflection, throttle and pitch angle) to reduce the cost function which represents forces and moments.

The ExitFlag = 1 indicates a successful convergence which shows that equilibrium has been reached. At higher speeds (160 and 180 m/s) there was a convergence failure which shows equilibrium was not reach due to thrust engine limits or insufficient conditions.

Task 3 Model linearization and eigenvalues:

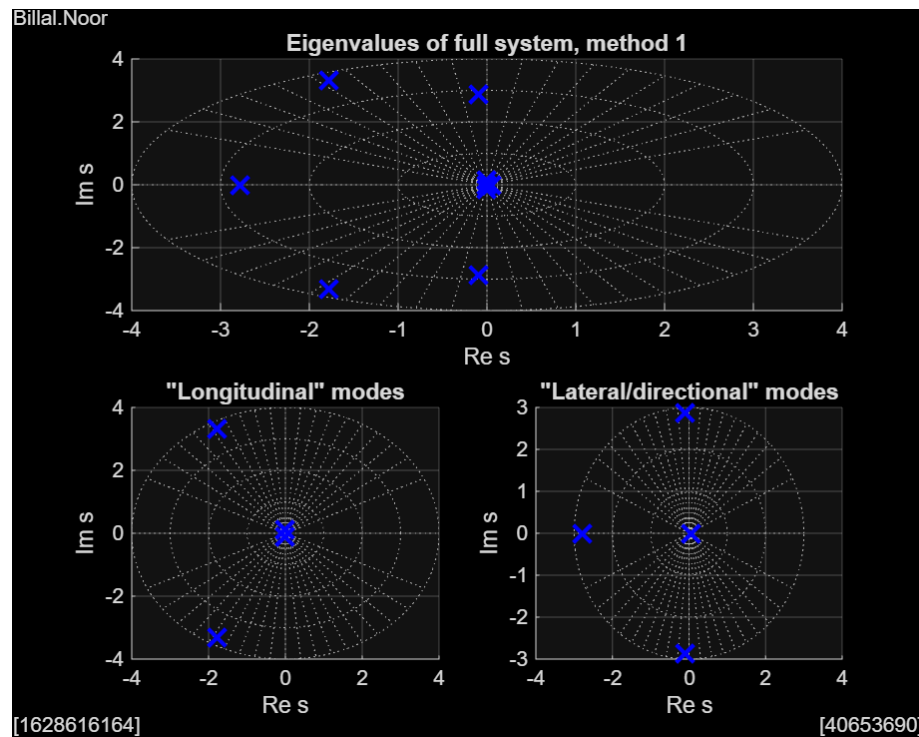


Figure 3: Eigenvalues and Longitudinal/Lateral modes

System matrix A

	1	2	3	4	5	6	7	8	9	10	11	12
1	-0.0245	0	0.0691	-0.0004	0.7970	0.0028	0	-9.8063	0	0	0	0
2	0	-0.2411	0	-2.5674	0	-146.8363	9.8063	0	0	0	0	0
3	-0.1404	-0.0050	-2.3471	-0.0827	148.4341	0.5234	-0.0002	0.0527	0.0002	0	0	-0.0010
4	0	-0.0981	0	-2.5436	0	2.6297	0	0	0	0	0	0
5	0.0046	-0.0045	-0.0769	-0.0750	-1.2148	0.4744	-0.0002	0	0.0002	0	0	0
6	0	0.0509	0	-0.1367	0	-0.1344	0	0	0	0	0	0
7	0	0	0	1.0000	0	-0.0054	0	0	0	0	0	0
8	0	0	0	0	1.0000	0	0	0	0	0	0	0
9	0	0	0	0	0	1.0000	0	0	0	0	0	0
10	1.0000	0	-0.0054	0	0	0	0	0	0	0	0	0
11	0	1.0000	0	0	0	0	0.8053	0	149.9975	0	0	0
12	0.0054	0	1.0000	0	0	0	0	-149.9975	0	0	0	0

System matrix B

	δe	δT	δa	δr
1	0.0212	0	0	3.2837
2	0	-0.3403	0.1827	0
3	-0.5774	-0.0020	0	0.0001
4	0	-0.5332	-0.2266	0
5	-0.1794	-0.0018	0	-0.2773
6	0	-0.0287	0.0779	0
7-12	0	0	0	0

Longitudinal subsystem

Matrix A

	1	2	3	4
1	-0.0245	0.0691	0.7970	-9.8063
2	-0.1404	-2.3471	148.4341	0.0527
3	0.0046	-0.0769	-1.2148	0
4	0	0	1.0000	0

Matrix B

	δe	δT
1	0.0212	3.2837
2	-0.5774	0.0001
3	-0.1794	-0.2773
4	0	0

Lateral subsystem

Matrix A

	1	2	3	4
1	-0.2411	-2.5674	-146.8363	9.8063
2	-0.0981	-2.5436	2.6297	0
3	0.0509	-0.1367	-0.1344	0
4	0	1.0000	-0.0054	0

Matrix B

	δa	δr
1	-0.3403	0.1827
2	-0.5332	-0.2266
3	-0.0287	0.0779
4	0	0

Eigenvalues of full system

Mode	Eigenvalue
1	$-1.7831 \pm 3.3306i$
2	$-0.0092 \pm 0.1274i$
3	$-0.0921 \pm 2.8646i$
4	-2.7863
5	-0.0018
6	+0.0514 (unstable)
Others	≈ 0

Longitudinal modes

Mode	Eigenvalue	ω_n	ζ
Short period	$-1.7829 \pm 3.3306i$	3.7778	0.4719
Phugoid	$-0.0103 \pm 0.1214i$	0.1218	0.0844

Lateral modes

Mode	Eigenvalue	ω_n / τ	ζ
Dutch roll	$-0.0921 \pm 2.8646i$	2.8661	0.0321
Roll subsidence	-2.7863	$\tau = 0.359$	—
Spiral	+0.0514	$\tau = 19.46$	—

The eigenvalues indicate that the longitudinal modes are stable with the short period mode being well damped and the phugoid mode lightly damped.

The lateral/directional dynamics show that the dutchroll mode is weakly damped and the roll subsidence mode is stable with fast decay. The spiral mode is unstable due to its positive real eigenvalue which shows divergence over time.

The longitudinal states are forward velocity, vertical velocity, pitch rate and pitch angle. The lateral states are side velocity, roll rate, yaw rate and roll angle.

The eigenvalues of the system matrix determine stability. If $\text{Re}(\lambda) < 0$ then you get stable (decaying response) and if $\text{Re}(\lambda) > 0$ then you get unstable (diverging response).

For oscillatory modes the natural frequency and damping ratio are calculated by using these equations

$$\omega_n = \sqrt{\sigma^2 + \omega^2} \quad \zeta = -\sigma / \omega_n \quad \text{where } \lambda = \sigma \pm i\omega$$

where $\lambda = \sigma \pm i\omega$

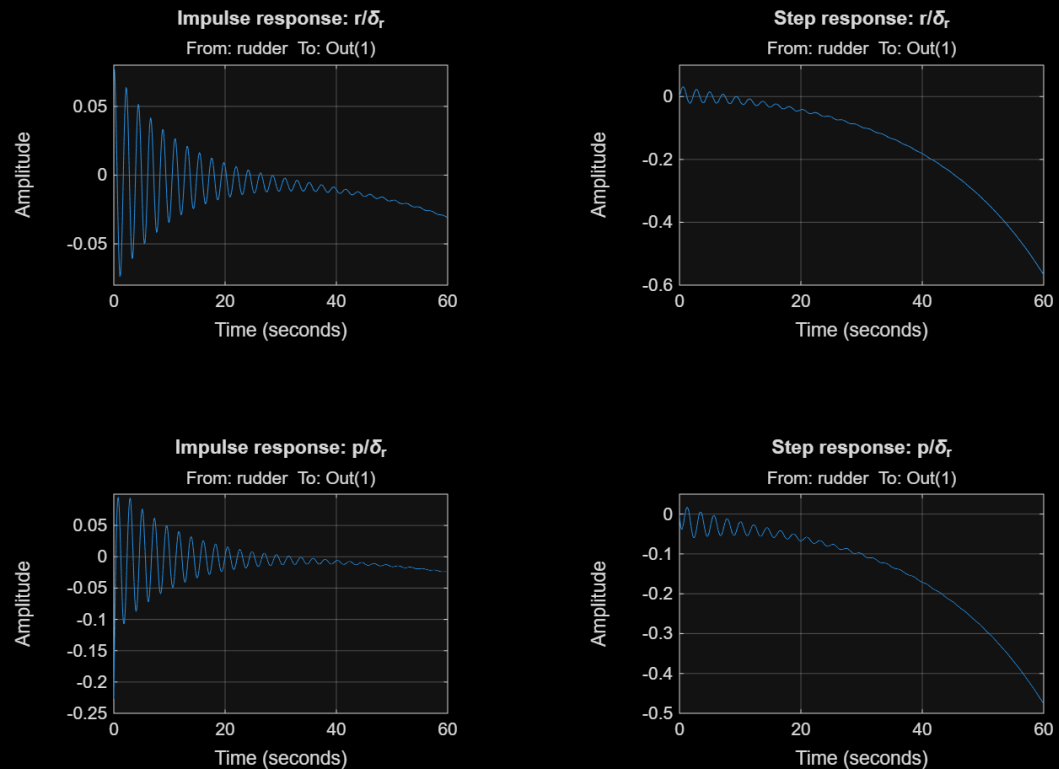
For real eigenvalues the time constant is

$$\tau = 1/|\lambda|$$

Task 4 transfer functions:

Transfer functions show the aircraft response for the elevator and rudder inputs which were obtained using the linearised model.

Elevator to Pitch rate (q)	$((-0.1794s^3 - 0.3809s^2 - 0.0107s) / (s^4 + 3.586s^3 + 14.36s^2 + 0.3466s + 0.2118))$
Elevator to Pitch angle (θ)	$((-0.1794s^2 - 0.3809s - 0.0107) / (s^4 + 3.586s^3 + 14.36s^2 + 0.3466s + 0.2118))$
Rudder to Yaw rate (r)	$((0.07787s^3 + 0.2571s^2 + 0.09137s - 0.03831) / (s^4 + 2.919s^3 + 8.575s^2 + 22.44s - 1.177))$
Rudder to Roll rate (p)	$((-0.2266s^3 + 0.1018s^2 - 0.5095s - 0.0002057) / (s^4 + 2.919s^3 + 8.575s^2 + 22.44s - 1.177))$



[1628616164]

[40653690]

Figure 4: Step and impulse response for the yaw and roll rate to rudder input.

The transfer functions show that the longitudinal dynamics are stable as the poles are located in the left half of the plane.

The MATLAB function `tf()` represents the system as a ratio of polynomials and `zpk()` is in terms of zeros poles and gain.

`tf()` is useful for algebraic manipulation calculations and `zpk()` makes system dynamics and stability more clearer to represent.

`Zpk()` is more useful for analysing stability since the poles are directly shown.

The results confirm that the longitudinal system is stable and the lateral system shows lightly damped oscillatory behaviour in the dutch roll mode

Task 5 Simulation:

The cessna 550 aircraft was simulated starting from the trim condition at 4000 m altitude and 150 m/s using different control inputs to excite specific flight modes.

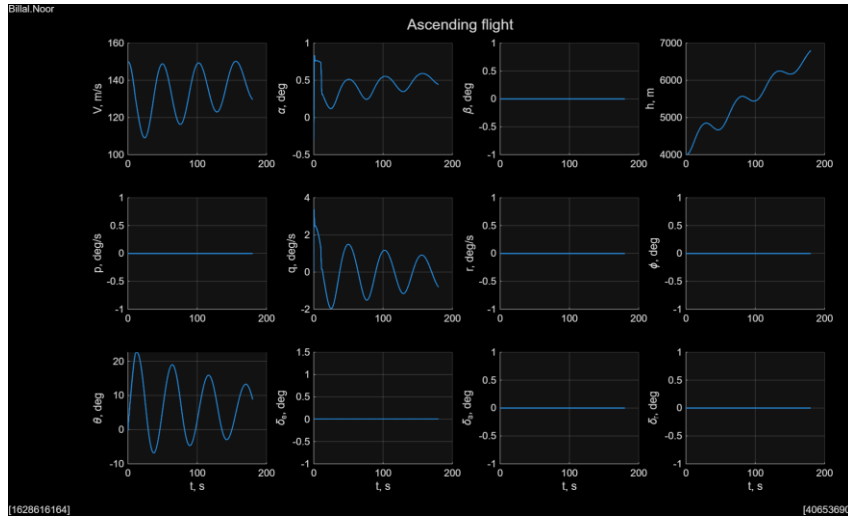


Figure 5: Aircraft response during ascending flight

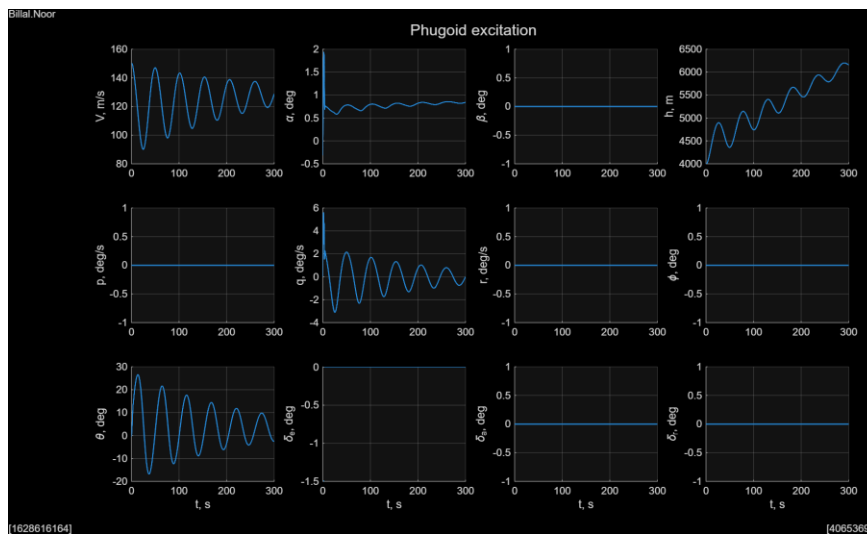


Figure 6: Phugoid mode response.

The phugoid shows slow oscillations with long periods meaning it is lightly damped.

The oscillation period (T) is measured from the plots and the natural frequency is calculated using $\omega = 2\pi/T$.

The values obtained correlate with the ones calculated from the eigen values which can show the model is linearised.

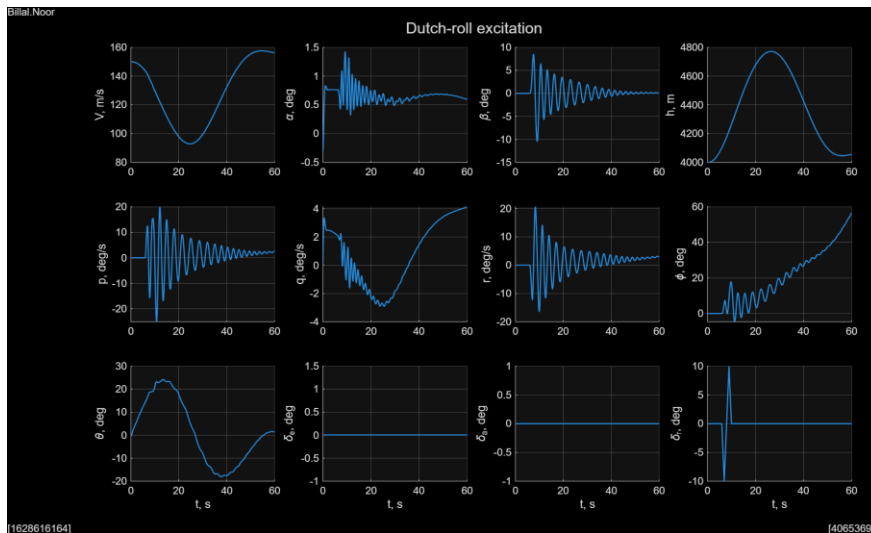


Figure 7:Dutch roll excitation.

The dutch roll mode exhibits oscillatory lateral motion with low damping which match the eigenvalue results.

The low damping ratio explains the oscillations in the graphs.

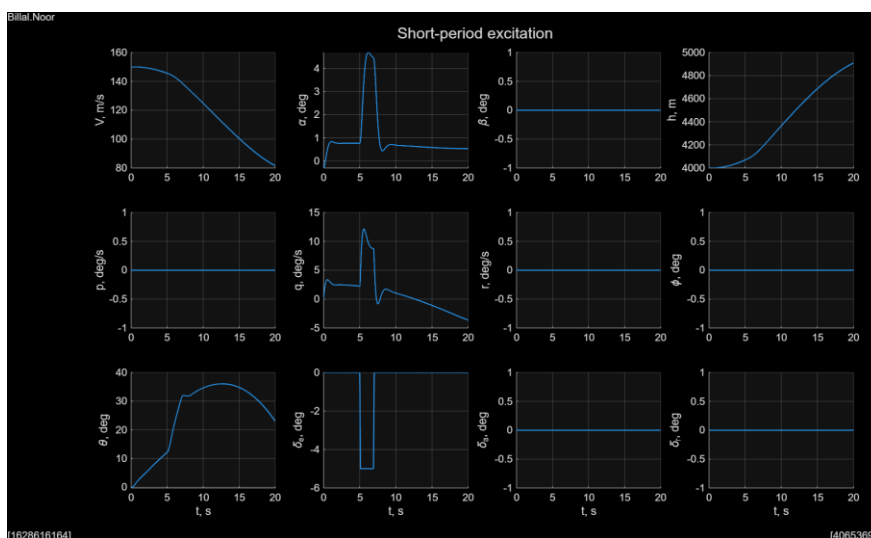


Figure 8:Short period mode response.

The short period mode shows fast oscillations that decay quickly which show good damping.

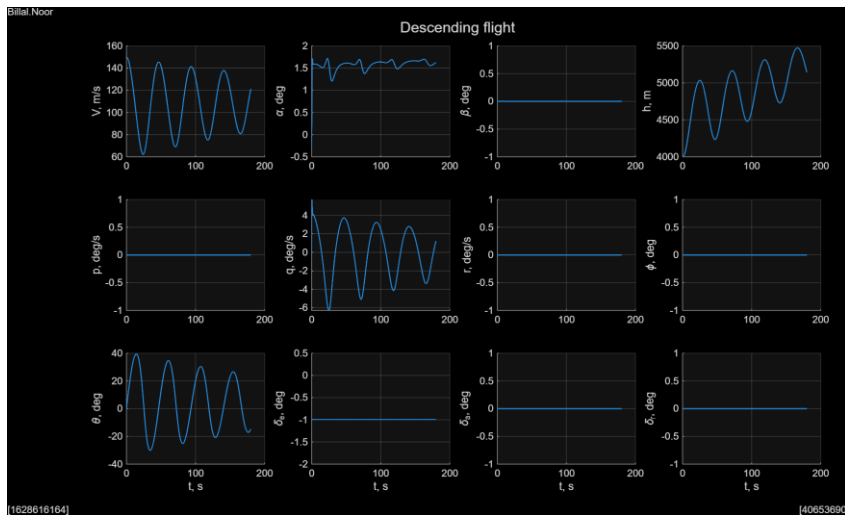


Figure 9:Aircraft response during ascending flight

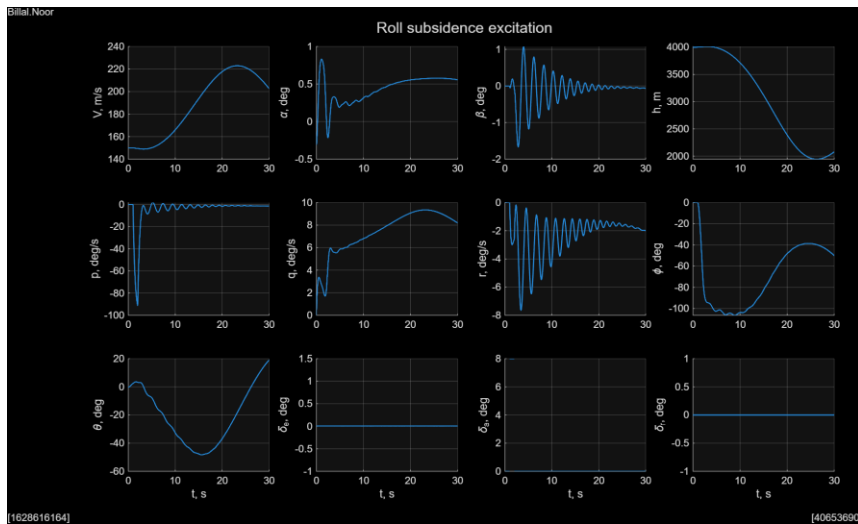


Figure 10:Roll subsidence mode.

The roll subsidence decays rapidly showing stable behaviour.

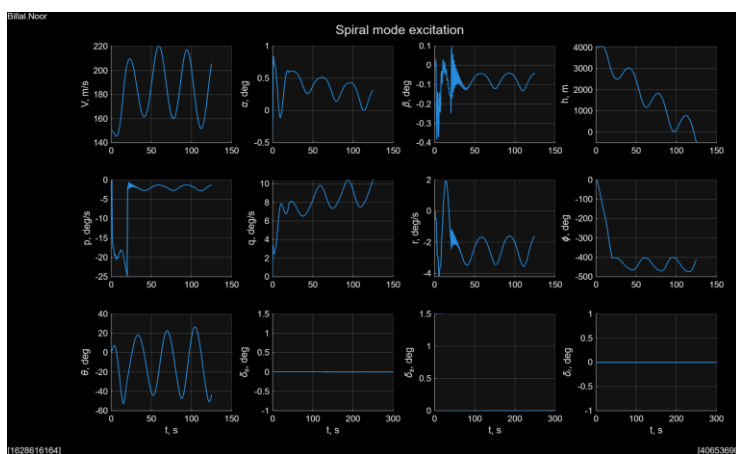


Figure 11: Spiral mode behaviour.

The spiral diverges over time confirming it is unstable. This confirms the positive real eigen value that was calculated and is consistent with the simulation results.

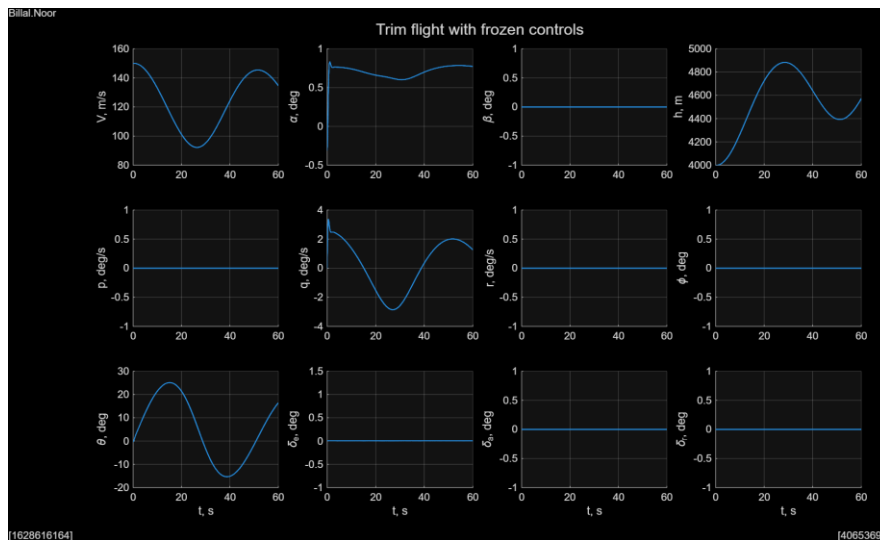


Figure 12: Aircraft response with fixed controls at trim condition.

With controls fixed at trim the aircraft remains approximately steady with small variations.

The oscillation periods observed in the simulation are consistent with the natural frequencies obtained from eigenvalues the small differences occur due to nonlinear effects in the full model.

The phugoid oscillations decay faster in descending flight compared to ascending flight. This is due to increased aerodynamic damping when the aircraft is descending. The increased aerodynamic forces lead to higher damping and cause decay and oscillations faster.

Task 6 Yaw damper:

The flying qualities of the aircraft show that the Dutch roll mode is lightly damped which can lead to oscillations.

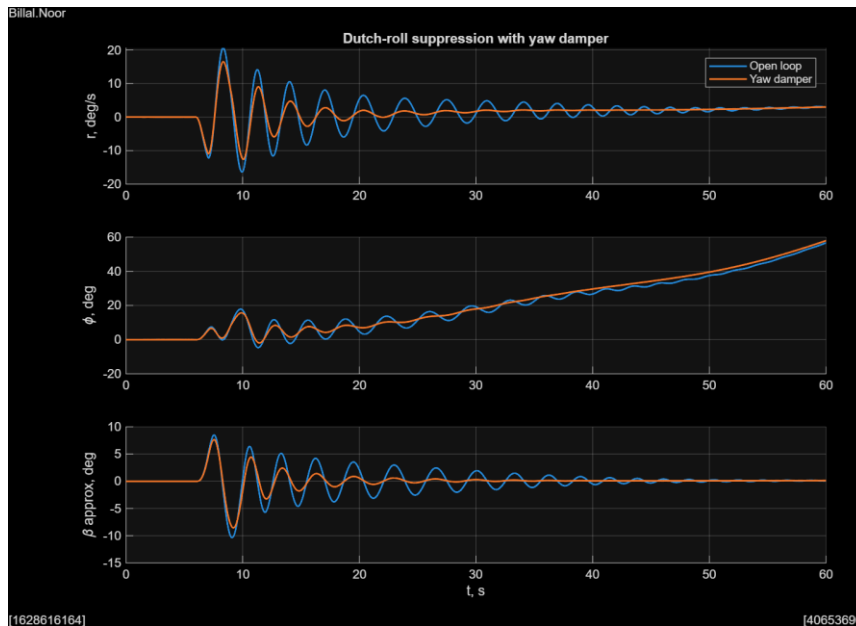


Figure 13: Comparison of aircraft response with yaw damper.

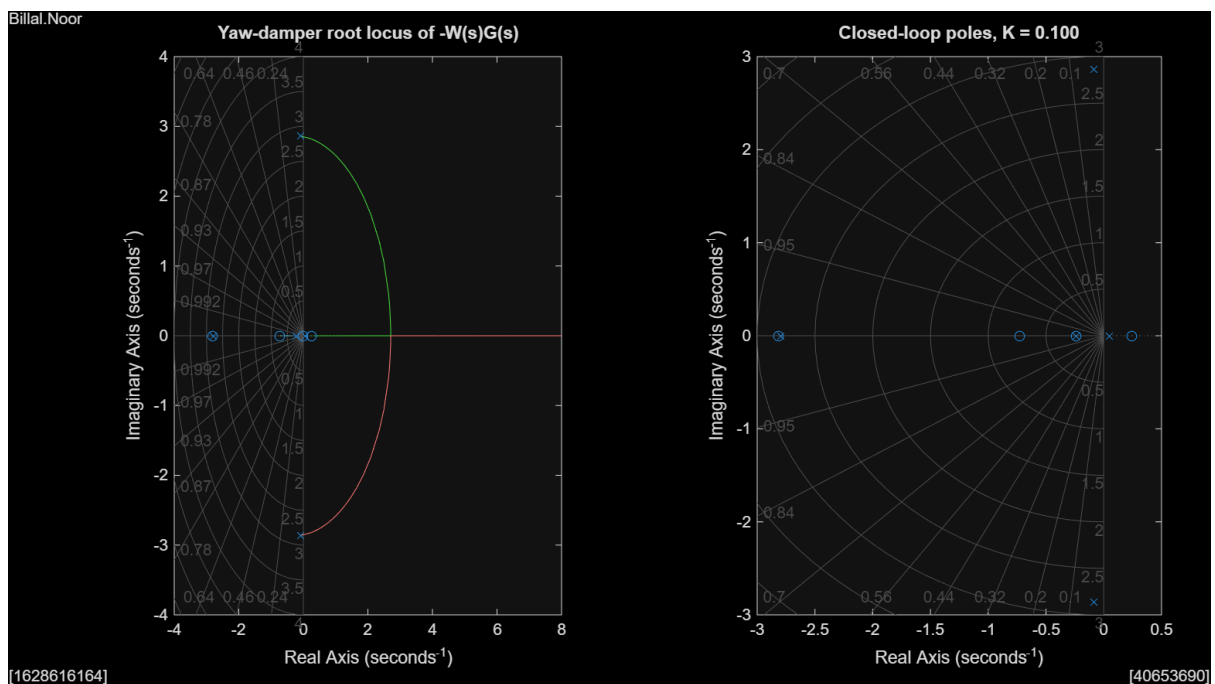


Figure 14: Yaw damper and closed loop imaginary/real comparison

A yaw damper was implemented using feedback from yaw rate:

$$\delta r = -K_r r \quad \delta r = -K_r r$$

where K_r is the feedback gain

The yaw damper significantly improves the damping of the dutch roll mode. The oscillations decay faster and the amplitude of motion is reduced.

Without the yaw damper the aircraft maintains oscillations due to low damping in the dutch roll mode. With the yaw damper the response becomes more stable and controlled.

This improves handling qualities and passenger comfort.